

Infrastructure Proof (Live Deployment)

This project provisions and manages a complete AWS infrastructure using **Terraform**, following real-world best practices (networking, load balancing, observability).

All components were **deployed, verified, and documented** in a real AWS environment.

Application Load Balancer

Public ALB routing traffic to EC2 instances behind an Auto Scaling Group.

ALB Access Logs in S3

ALB access logs are enabled and stored in a dedicated S3 bucket for auditing and troubleshooting.

CloudWatch Metrics

Infrastructure metrics (ALB and EC2) are available in CloudWatch for monitoring and observability.

Terraform Deployment

Infrastructure fully created using Terraform.

Reproducibility

The infrastructure can be reproduced from scratch using:

```
cd envs/dev  
terraform init  
terraform apply
```

All resources are managed exclusively via Terraform (no manual changes).

Why this works

- Real AWS deployment (not theoretical)
 - Observability enabled (logs + metrics)
 - Infrastructure as Code (Terraform)
 - Clean and reproducible setup
-

Notes

- Screenshots are taken from a real AWS account
 - Sensitive data has been intentionally hidden
 - Resources were destroyed after validation to avoid unnecessary costs
-

Result

This repository demonstrates **practical cloud infrastructure skills**, not just configuration files

```
● usuario1@vbox:~/Documents/Terraform/proyecto1/envs/dev$ terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/tls from the dependency lock file
- Using previously-installed hashicorp/tls v4.1.0
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
module.rds.aws_db_instance.this: Still creating... [04m40s elapsed]
module.rds.aws_db_instance.this: Still creating... [04m50s elapsed]
module.rds.aws_db_instance.this: Still creating... [05m00s elapsed]
module.rds.aws_db_instance.this: Creation complete after 5m10s [id=db-I2ZED3R56BG4EMDILKDZMYX67E]
```

Apply complete! Resources: 91 added, 0 changed, 0 destroyed.

Outputs:

```
rds_endpoint = "postgres-dev.chc8is0s19e2.us-west-2.rds.amazonaws.com:5432"
rds_port = 5432
s3_bucket_arn = "arn:aws:s3:::project-dev-logs-luisrod"
s3_bucket_name = "project-dev-logs-luisrod"
usuario1@vbox: ~/Documents/Terraform/proyecto1/envs/dev$
```

VPCs ► See all regions	Oregon 1	Endpoint Services ► See all regions	Oregon 0
Subnets ► See all regions	Oregon 4	NAT Gateways ► See all regions	Oregon 0
Route Tables ► See all regions	Oregon 2	VPC Peering Connections ► See all regions	Oregon 0
Internet Gateways ► See all regions	Oregon 1	Network ACLs ► See all regions	Oregon 1
Egress-only Internet Gateways ► See all regions	Oregon 0	Security Groups ► See all regions	Oregon 4
DHCP option sets	Oregon 1	Customer Gateways	Oregon 0

```
module.static_site_s3.aws_s3_object.static_files["fontawesome/webfonts/fa-regular-400.woff2"]: Creation complete after 0s [id=mi-bucket-luis-fernando/fontawesome/webfonts/fa-regular-400.woff2]
module.static_site_s3.aws_s3_object.static_files["img/img-16.jpg"]: Creating...
module.static_site_s3.aws_s3_object.static_files["index.html"]: Creation complete after 0s [id=mi-bucket-luis-fernando/index.html]
module.static_site_s3.aws_s3_object.static_files["fontawesome/webfonts/fa-solid-900.eot"]: Creating...
module.static_site_s3.aws_s3_object.static_files["fontawesome/webfonts/fa-brands-400.woff2"]: Creation complete after 1s [id=mi-bucket-luis-fernando/fontawesome/webfonts/fa-brands-400.woff2]
module.static_site_s3.aws_s3_object.static_files["img/hero.jpg"]: Creating...
module.static_site_s3.aws_s3_object.static_files["video-detail.html"]: Creation complete after 1s [id=mi-bucket-luis-fernando/video-detail.html]
module.static_site_s3.aws_s3_object.static_files["img/img-01.jpg"]: Creating...
module.compute.aws_iam_instance_profile.ec2_static_profile: Creation complete after 7s [id=ec2-static-site-profile]
module.static_site_s3.aws_s3_object.static_files["fontawesome/attribution.js"]: Creating...
module.s3.aws_s3_bucket_versioning.this: Creation complete after 3s [id=project-dev-logs-luisrod]
module.static_site_s3.aws_s3_object.static_files["img/img-14.jpg"]: Creating...
module.static_site_s3.aws_s3_object.static_files["fontawesome/LICENSE.txt"]: Creation complete after 1s [id=mi-bucket-luis-fernando/fontawesome/LICENSE.txt]
module.static_site_s3.aws_s3_object.static_files["img/img-02.jpg"]: Creating...
```

```
usuario1@vbox: ~/Documents/Terraform/proyecto1/envs/dev$ terraform state list
aws_ssm_parameter.db_password
aws_ssm_parameter.db_username
module.alb.aws_lb.this
module.alb.aws_lb_listener.http
module.alb.aws_lb_target_group.web_tg
module.alb.aws_security_group.alb_sg
module.asg.aws_autoscaling_group.this
module.asg.aws_launch_template.this
module.cloudwatch_ec2.aws_cloudwatch_log_group.this
module.compute.data.aws_ami.compute_amazon_linux
module.compute.aws_iam_instance_profile.ec2_static_profile
module.compute.aws_iam_policy.static_s3_read
module.compute.aws_iam_role.ec2_static_role
module.compute.aws_iam_role_policy_attachment.attach_static_policy
module.compute.aws_instance.compute_spot
module.compute.aws_key_pair.compute_key
module.compute.aws_lb_target_group_attachment.this
module.compute.aws_security_group.compute_sg
module.compute.tls_private_key.compute_ssh
module.networking.aws_internet_gateway.this
module.networking.aws_route_table.public
module.networking.aws_route_table_association.public[0]
module.networking.aws_route_table_association.public[1]
module.networking.aws_subnet.private[0]
```

Instances (running)	2	Auto Scaling Groups	1	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	2	Instances	2
Key pairs	5	Load balancers	1	Placement groups	0
Security groups	4	Snapshots	0	Volumes	2

☐	asg-spot-web	i-02a5b470d6b4d0a5e	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2b
☐	compute-spot-web	i-02d35d9ff48d6a04e	Running	t3.micro	3/3 checks passed	View alarms +	us-west-2a

Instances (1/2) Info									
Find Instance by attribute or tag (case-sensitive)									
All states									
Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	MI	
micro	3/3 checks passed	View alarms +	us-west-2b	ec2-35-162-220-111.us...	35.162.220.111	-	-	dis	
micro	3/3 checks passed	View alarms +	us-west-2a	ec2-44-250-252-255.us...	44.250.252.255	-	-	dis	

web-alb-dev-tg

Details

arn:aws:elasticloadbalancing:us-west-2:459566754179:targetgroup/web-alb-dev-tg/064bb6d0d964dd5f

Target type
Instance

Protocol : Port
HTTP: 80

Protocol version
HTTP1

VPC
[vpc-07752a69eeb156130](#)

IP address type
IPv4

Load balancer
[web-alb-dev](#)

2
Total targets

2 Healthy

0 Unhealthy

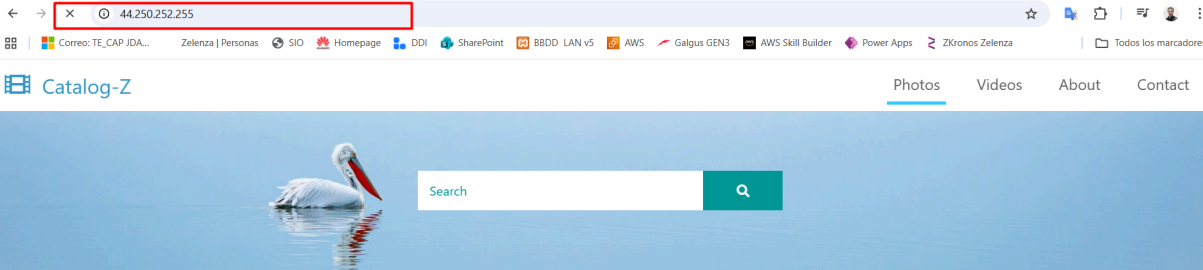
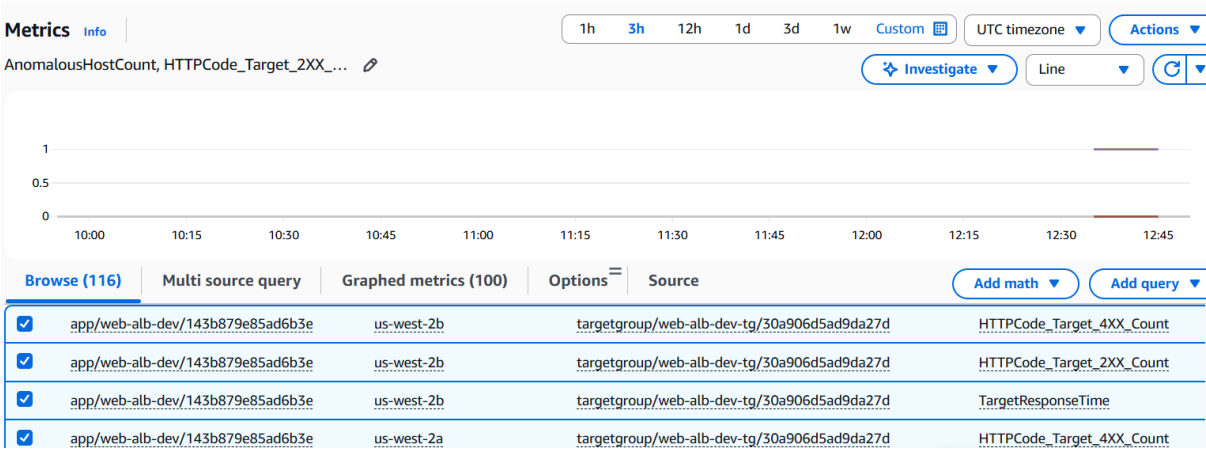
0 Unused

0 Initial

0 Draining

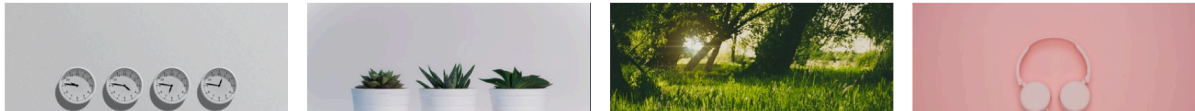
Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.



Latest Photos

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AWS estimated bill summary Info

Total charges and payment information

Account ID: 459566754179 | Billing period: December 1 - December 31, 2025 | Bill status: Pending

Service provider: Amazon Web Services EMEA SARL | Total in USD: USD 0.81

Estimated grand total: USD 0.81

Payment information Info

Highest estimated cost by service provider Info

Viewing Amazon Web Services EMEA SARL

Highest service spend: USD 0.49 | Trend compared to prior month: ↓ 84.1%

Highest AWS Region spend: USD 0.67 | Region name: US West (Oregon)

Service name: Virtual Private Cloud

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```
Plan: 91 to add, 0 to change, 0 to destroy.
```

```
Changes to Outputs:
```

```
+ rds_endpoint = (known after apply)
+ rds_port     = (known after apply)
+ s3_bucket_arn = (known after apply)
+ s3_bucket_name = "project-dev-logs-luisrod"
```

```
Do you want to perform these actions?
```

```
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
```

```
Enter a value: yes
```

```
module.compute.tls_private_key.compute_ssh: Creating...
module.networking.aws_vpc.this: Creating...
module.compute.aws_iam_role.ec2_static_role: Creating...
```

```
module.networking.aws_internet_gateway.this: Still destroying... [id=igw-0cf87531ffd142040, 05m00s elapsed]
module.asg.aws_autoscaling_group.this: Still destroying... [id=asg-spot-web, 05m10s elapsed]
module.networking.aws_internet_gateway.this: Still destroying... [id=igw-0cf87531ffd142040, 05m10s elapsed]
module.asg.aws_autoscaling_group.this: Still destroying... [id=asg-spot-web, 05m20s elapsed]
module.networking.aws_internet_gateway.this: Still destroying... [id=igw-0cf87531ffd142040, 05m20s elapsed]
module.asg.aws_autoscaling_group.this: Still destroying... [id=asg-spot-web, 05m30s elapsed]
module.networking.aws_internet_gateway.this: Still destroying... [id=igw-0cf87531ffd142040, 05m30s elapsed]
module.asg.aws_autoscaling_group.this: Still destroying... [id=asg-spot-web, 05m40s elapsed]
module.networking.aws_internet_gateway.this: Still destroying... [id=igw-0cf87531ffd142040, 05m40s elapsed]
```

```
module.asg.aws_launch_template.this: Destruction complete after 1s
module.static_site_s3.aws_s3_bucket.static_site: Destroying... [id=mi-bucket-luis-fernando]
module.alb.aws_security_group.alb_sg: Destroying... [id=sg-0ec34b18106a41661]
module.alb.aws_lb_target_group.web_tg: Destruction complete after 1s
module.networking.aws_subnet.public[0]: Destruction complete after 1s
module.networking.aws_subnet.public[1]: Destruction complete after 2s
module.alb.aws_security_group.alb_sg: Destruction complete after 1s
module.static_site_s3.aws_s3_bucket.static_site: Destruction complete after 1s
module.networking.aws_vpc.this: Destroying... [id=vpc-07752a69eeb156130]
module.networking.aws_vpc.this: Destruction complete after 1s
```

```
Destroy complete! Resources: 91 destroyed.
```

```
usuario1@vbox: ~/Documents/Terraform/proyecto1/envs/dev$
```